Static and kinetic friction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish adjustable static friction from sliding friction.

SUCCESS CRITERIA - I can...

1. Compare a push with the static-friction limit.
2. Find actual friction for rest or sliding.
3. Explain friction direction using relative slipping.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “friction” and “static” to compare your model or result. Sentence starters are optional.

friction: a contact force opposing relative sliding or its tendency

static: not sliding relative to the contacting surface

kinetic: associated with motion; kinetic friction acts during sliding

01 Bring back an earlier idea

A spacecraft of mass 3 kg pushes an asteroid of mass 300 kg with 15 N.
Compare interaction forces and the accelerations due to that interaction.

Word help: interaction = two objects affecting one another

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $0 \le f_s \le \mu_s N$; $f_k = \mu_k N$.
s = static; k = kinetic; N = normal force. Static friction adjusts up to its maximum. Kinetic friction applies during sliding.

Worked reasoning: A box remains still while pushed with 3 N. Static friction is 3 N opposite the push, provided its maximum is at least 3 N. Static friction need not equal its maximum.

Your turn: Draw the horizontal forces. Then redraw for a 5 N push that is still below the maximum.

Sentence starter: The … in my model represents … . Its label shows … .

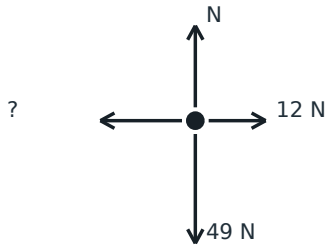
Marzano check - circle: ① ② ③

03 Actual versus maximum

A 5 kg box on a level floor has $\mu_s = .40$ and $\mu_k = .30$.
Horizontal push 12 N; initially at rest.
Find N, f_s, max, actual friction and a.
Use g=9.8.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.



The dot represents the object. Initially at rest

Marzano check - circle: ① ② ③

04 Sliding case

Increase the push on that box to 25 N.
Find friction and acceleration after it slides.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Estimate from data

Supplied trials for normal force 20 N: peak forces 8, 9, 7 N; steady sliding forces 6, 5, 7 N.
Find mean peak, mean sliding force, and coefficient estimates. Which coefficient describes threshold?

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

06 Choose a method

x_i=-4±0.5 m, v=6±0.1 m/s and exact t=8 s.
Give central final position, its conservative interval and full width.
Name an assumption.

Word help: final = at the end of the interval; subscript f

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

A 10 kg box hasμ_s=.50,μ_k=.40, g=9.8.
Compare 30 N and 60 N horizontal pushes, starting from rest each time.
Give actual friction and acceleration.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 5 kg box has μ_s=0.4 and μ_k=0.3 on a level floor; use g=10 N/kg. A horizontal 22 N pull begins from rest.
Find maximum static friction, acceleration after motion starts, and the pull needed for steady sliding.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3